

TECHNICAL MANUAL
FOR
KIRLOSKAR PUMP
TYPE DB

KIRLOSKAR BROTHERS LIMITED
UDYOG BHAVAN, TILAK ROAD, PUNE- 411002

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0 MODELS

'DB' types of pumps are available in following models:

32/13	40/13	50/13	65/13	80/16	100/20	125/26	150/32
32/16	40/16	50/16	65/16	80/20	100/26	125/32	150/40
32/20	40/20	50/20	65/20	80/26	100/32	125/40	
32/26	40/26	50/26	65/26	80/32	100/40		
		50/32K	65/32				

1 APPLICATIONS

'DB' pumps are used for cold and hot water and non- corrosive chemicals. The fields of application of these pumps include water works for mechanically Filtered water, for circulation of water in air conditioning plants, textile mills, Industrial plants, for pumping, screening water for paper and pulp mills, for Lift irrigation schemes and agricultural purposes, in sugar industry for pumping clear and weighted juice, with small fire fighting equipment etc.

For any other application, please refer to our sales head office.

2 GENERAL DESCRIPTION

2.1 Constructional features:

All 'DB' type of pumps except 32/26 are constructed as standard water pumps in accordance with DIN 24255 as far as their main dimensions and performance ratings are concerned. These are single stage, single suction horizontal shaft, volute type foot mounted pumps with the bearing bracket also having a support foot. Mounted pumps with the bearing bracket also having a support foot. These are centrifugal pumps with enclosed type Impeller and are fitted with standard accessories.

2.2 Liquid conditions

These pumps are suitable for clean liquids not contaminated by any solid particles. Maximum temperature of the pumped liquid is as below:

Shaft seal	max. Temp. of pumped liquid in ° c
1) Stuffing box without cooling	-10 to +100

2.3 Special features

The special feature of these pumps is that they are manufactured on the module construction so that a number of individual components will fit several pump sizes and are interchangeable between pump sizes. They also offer back pull out design with spacer type coupling which allows access to the pump interior for maintenance purpose without disturbing the pipe work or removal of prime mover. This advantage reduces service time resulting in lower maintenance costs and reduction in production losses.

3 OPERATING DATA

3.1 Capacity of water fill.

The capacity of water fill for 'DB' pumps is as follows:

Size	Water fill in liters	Size	Water fill in liters
32/13	0.8	65/26	3.9
32/16	1.0	65/32	7.3
32/20	1.2	80/16	4.0
32/26	1.8	80/20	4.4
40/13	1.1	80/26	4.9
40/16	1.2	80/32	8.3
40/20	1.5	100/20	6.5
40/26	2.1	100/26	8.0
50/13	1.8	100/32	9.6
50/16	1.8	100/40	13.8
50/20	2.1	125/26	12.0
50/26	3.9	125/32	14.0
50/32 K	6.9	125/40	15.5
65/13	2.4	150/32	20.0
65/16	2.8	150/40	24.0
65/20	3.1		

3.2 Rotational speeds.

The maximum speeds for different sizes are

32/13 to 65/13	}	$n \leq 3500$ rpm
32/16 to 80/16		
32/20 to 100/20		
32/26 to 100/26	}	for impeller of C.I.
		$n \leq 2900$ rpm at full imp. dia.
		For Diameter = 218 mm * or below
		$n \leq 3500$ rpm
125/26	}	$n \leq 1700$ rpm
100/32 to 150/32		
100/40 to 150/40		
50/32 to 80/32		

Note: Restrictions due to maximum peripheral speed on impellers of C. I. (40 m/s) the minimum rotational speed for all sizes are 700 rpm.

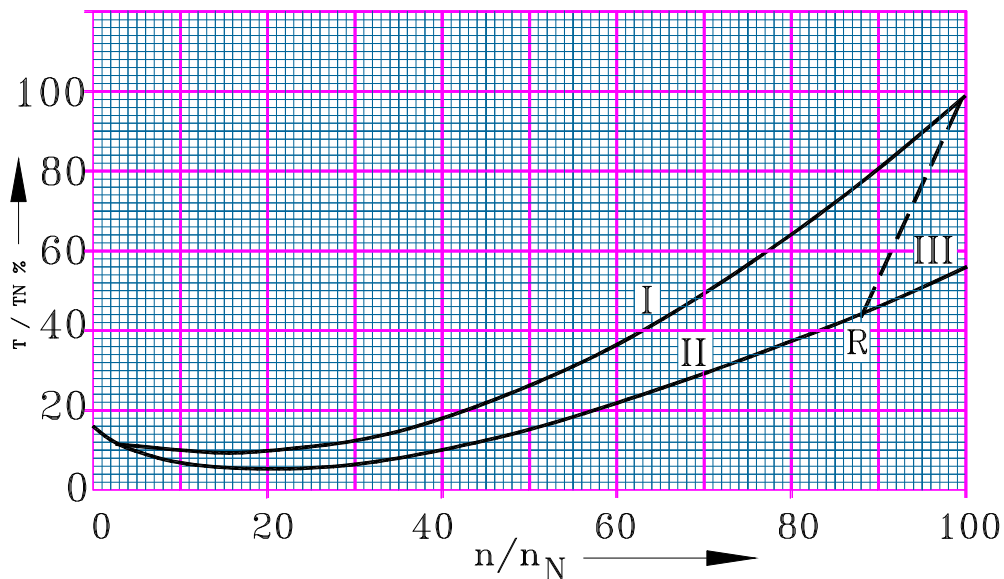
3.3 maximum casing hydro test pressures

Size								18.7Kg/cm ²
32/13	40/13	50/13	65/13	80/16	100/20	125/26	150/32	
32/16	40/16	50/16	65/16	80/20	100/26	125/32	150/40	
32/20	40/20	50/20	65/20	80/26	100/32	125/40		
32/26	40/26	50/26	65/26	80/32	100/40			
		50/32K	65/32					

3.4 Net Positive Suction Head

The printed NPSH curves are based on actual valves of NPSH achieved on our test bed. A minimum safety margin of 0.5 meters must be added to this when applying the same to the site conditions.

3.5 Torque speed curve



- Case 1 Stating up with open isolating or regulating valve on delivery side
- Case 2 Starting up with a closed isolating or regulating valve with opening the valves when nominal speed is attained.
- Case 3 Starting up with an open isolating or regulating valve but against a static delivery head static on a non return valve.

Note: Case 2 is applicable to DB pumps however; motor should be designed and suitable for all the 3 cases

Starting – up with open isolating or regulating valve on delivery

From the pump curves (whilst considering the sp. gravity of the pumped liquid), we can take:

Nominal speed } N in rpm for the operating conditions as per contract
 Nominal power required } P_N in kW

The nominal moment of the pump amounts to $M_{pN} = P_N / n_N$ 974 in Kpm

For any speeds between $n = 0$ and n_N the starting moment M_P of the pump is then

$$M_p = (n/n_N)^2 M_{PN}$$

Plotting of the starting moment curve:

From the curve 1 we can read ratings M/M_N in % for any ratings n/n_N in %

From that follows

$$M_P = (M/M_N) \text{ as per curve 1. } \frac{M_P N}{100} \text{ in kpm}$$

EXAMPLE:

$$n_N = 2900 \text{ 1/min}$$

$$M_{PN} = \frac{974 * 51}{2900} = 33.5 \text{ Kpm}$$

$$P_N = 8 \text{ KW}$$

n in rpm	n/n _N in %	M/ M _m in %	M _p in kpm
725	25	10	0.27
1440	50	25	0.68

Case 2: starting –up with a closed isolating or regulating valve, with opening the valves when the nominal speed is attained

From the pump curves (whilst considering the sp. gravity of the pumped liquid), we can take:

Nominal speed n_N in rpm

Best performances p_{opt} in kW

for the point of best efficiency with the appropriate impeller dia. for the operation conditions.

Momentum of the pump for the best efficiency point is then $m_{popt} = 974 \frac{prot}{n_N}$ in kpm

Plotting of the starting moment curve:

From the curve 2 if we can read ratings M/M_N in % for any ratings n/n_N in % from that follows

$$M_P = (M/M_N) \text{ as per the curve 2 } M_{prot} \text{ in kpm}$$

$$\text{Example: } n_N = 1480 \text{ 1/min } m_{popt} = (974 * 51) / 1480 = 33.5 \text{ kpm}$$

$$p_{popt} = 51 \text{ kw}$$

N in 1/min	n/n _N in %	M/M _N in %	M _p in kpm
0	0	15	5.0
370	25	5	1.7
740	50	15	6.0

3.6 Minimum safe flow

The minimum safe flow valves are based on temperature rise at lower discharges and performance at 2900 rpm. Pump types * marked are at 1450 rpm.

Size	Min. safe flow in litre per sec	Size	Min. safe flow. in litre per sec
32/13	0.75	65/26	4.0
32/16	0.75	65/32 *	4.0
32/20	0.75	80/16	4.0
32/26	0.75	80/20	4.0
40/13	0.75	80/26	4.0
40/16	1.5	80/32	4.0
40/20	1.5	100/20	4.0
40/26	1.5	100/26	5.6
50/13	2.0	100/32 *	5.25
50/16	2.0	100/40 *	4.0
50/20	2.0	125/26 *	5.3
50/26	2.0	125/32 *	4.0
50/32K	4.0	125/40 *	4.0
65/13	3.0	150/32 *	7.0
65/16	3.0	150/40 *	8.0
65/20	4.0		

4.0 CONSTRUCTION

4.1 Casing

The casing for 'DB' pumps is of closed grained, high tensile C.I. volute type.

Suction and discharge nozzles as well as the supporting feet are cast integral with the casing.

4.1.1 Flanges

All companion flanges are drilled to DIN 2533 ND16 PLAIN FACE.

Models	Suc. bore in mm	Del. bore in mm
32/13, 32/16 32/20, 32/26	50	32
40/13, 40/16 40/20, 40/26	65	40
50/13, 50/16 50/20, 50/26, 50/32K	65	50
65/13, 65/16 65/20, 65/26, 65/32	80	65
80/16, 80/20 80/26, 80/32	100	80
100/20, 100/26 100/32, 100/40	125	100
125/26, 125/32 125/40	150	125
150/32, 150/40	200	150

Note: Drilling of flanges as per BS or ASA or any other standards will be offered at extra cost.

4.1.2 Nozzle orientation

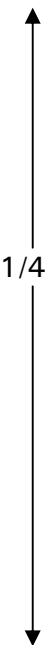
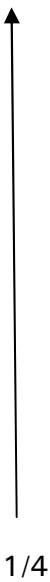
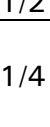
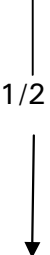
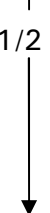
For 'DB' pumps, the suction branch is axial while the delivery branch is radial and is vertically upwards.

4.1.3 Casing tapings

The casing tapping are provided for

- 1) Vacuum gauge TP01
- 2) Pressure gauge TP02
- 3) Delivery casing drain TP03
- 4) Stuffing box leakage drain TP04
- 5) Stuffing box sealing TP05 (external)
- 6) Venting plug TP010

Casing tapping details in inch BSP are as follows

Size	TP 01	TP 02	TP03	TP04	TP05	TP10	
32/13							
32/16							
32/20							
32/26,							
40/13							
40/16							
40/20							
40/26							
50/13							
50/16							
50/20							
50/26							
50/32K,							
65/13							
65/16							
65/20							
65/26							
65/32							
80/16							
80/20							
80/26							
80/32							
100/20							
100/26							
100/32							
100/40							
125/26							
125/32							
125/40,							
150/32							
150/40							

4.1.4 Casing rings

Each pump is provided with a casing ring on suction side to form a seal against leakage from the discharge chamber back to the suction, and replacement in case of wear. In case of large size pump casing ring on the back side of the impeller is also provided. They are made of C.I. (IS 210 Gr FG 260) or bronze LT. Br. IS 318 Gr L T B 2 OR Phosper bronze (IS 28 Gr 1)

Size	Inside dia. in mm		Outside in mm	
	Suction	Delivery	Suction	Delivery
32/13	70	*	80	*
32/16	70	*	80	*
32/20	80	*	90	*
32/26	80	*	90	*
40/13	80	*	90	*
40/16	80	*	90	*
40/20	80	*	90	*
40/26	95	*	110	*
50/13	95	*	110	*
50/16	95	*	110	*
50/20	95	*	110	*
50/26	115	*	130	*
50/32K	115	125	130	170
65/13	115	*	130	*
65/16	115	*	130	*
65/20	115	*	130	*
65/26	115	*	130	*
65/32*	130	125	145	170
80/16	130	*	145	*
80/20	130	*	145	*
80/26	130	*	145	*
80/32	140	135	158	170
100/20	160	*	175	*
100/26	160	*	175	*
100/32	160	155	175	170
100/40	160	160	175	220
125/26	180	*	200	*
125/32	180	180	200	220
125/40	180	180	200	220
150/32	200	200	220	220
150/40	200	200	220	220

* No casing ring is provided for these sizes on delivery side as these impellers are with back vanes

4.2 Impellers

Impellers of these pumps are of closed grained C.I. (IS 210 gr FG 260) or bronze (LT Br IS 318 Gr LT B 2) or phosphor bronze (IS 28 Gr 1) they are single suction, enclosed type and accurately balanced dynamically. The impeller is hydraulically balanced by back vanes and in case of large impeller diameters by balancing holes.

4.2.1 Thrust balancing

For 'DB' pumps, Axial thrust balancing is done as under	
Models	Type of balancing
32/13 to 65/13 32/16 to 80/16 32/20 to 100/20 32/26 to 125/26	back vanes for Impellers
50/32K to 150/32 100/40 to 150/40	relief holes and wear ring on delivery side

4.3 Bearing and lubrication

4.3.1 Bearing

'DB' pumps are provided with antifriction heavy-duty ball bearing, which are capable of carrying large thrust loads in either direction. These bearing need only proper lubrication, wear is negligible resulting in accurate alignment during many years of service.

4.3.1.1 Bearing details

Size	NDE Bearing SKF or Equi.	DE Bearing SKF or Equi.	Grease content gms.	Oil content of bearing housing
32/13, 32/16,32/20,32/26 40/13,40/16,40/20,40/26 50/13,50/16,50/20,50/26 65/13,65/16,65/20, 80/16.	6305	6305	5.0	0.2
65/26,65/32 80/20,80/26,80/32 100/20,100/26,100/32 125/26	6307	6307	8.0	0.55
100/40 125/32,125/40 150/32,150/40	6309	6309	12.0	0.9

Grease and oil content values given are approximate.

4.3.2 Lubrication:

Grease lubricated bearing will be the standard supply. Pumps with oil-lubricated bearings will be supplied against specific order. The details of grease/oil used are as under.

Grease lubricated bearings should be repacked with a high quality lithium soap base ball and roller bearing grease, free from resin and acid not liable to harden or

crumble, and possessing rust preventive properties. The grease should have a penetration no. situated between 2 to 3, corresponding to a worked penetration of 220 to 295 mm/10. The drop point should not be lower than 150°C (300°F) in view of the above, the recommended lubricant specifications for 'DB' pumps are as under:

- a) Type of lubricant : grease
- b) Specifications of lubricant : Caltax Regal Star FAK No. 2 or equivalent grease
Viscosity 360 SSU at about 38 °C (100°F)
Dropping point (150°C) (300°F)
- c) Period of lubrication : Approx. after every working 1000 running hours
- d) Quality of lubricant : As shown in the table above

We also recommend the following greases:

ARAL	ARAL Fett HL 3
BP	BP ENERGREASE LS 3
CHEVRON	CHEVRON Dura –Lith grease 3
DEA	DEA VISCOBIL Fett FT 42
MOBIL	MOBILUX 3
SHELL	SHELL Alvania fett 3
VALVOLINE	VALVOLINE LB -2
IOC	SERVOGEM 3 upto 1500 rpm
	SERVOGEM 2 above 1500 rpm

The bearings are greased for the first time in the factory.

Re-charge the bearings, using a greased gun, through the two conical grease nipples.

Re-greasing intervals for pump speeds-

- 1500 rpm about every 5000 operating hours
- 3000 rpm about every 2500 operating hours

After about 10,000 operating hours, not later than two years, however dismantle the bearings, wash with petrol, dry then and furnish them with fresh filling grease.

When service conditions are particularly unfavorable (dusty and wet service, high ambient temperature), then intervals for re-greasing must be corresponding shorter.

Re-greasing rates :

The amount of grease required for re-charging is determined by the size of the bearing and is already given in 4.3.1.1

Note : if the bearings are over greased (too much grease in the bearings), there is a danger of hot running.

Variation in lubrication:

The pumps are supplied with oil-lubricated ball bearings. If the bearing temperature is always below 50°C and if there is only small risk of contamination, oil should be changed about once a year. If the bearing temperature is upto 80°C and has a danger of contamination. Oil should be changed about every six months.

All the bearings have slackness C 3

Cooling of the oil in bearing housing is not possible.

Variation: Oil lubrication instead of grease.

The following oils are recommended for the lubrication of the bearings.

To be used for	Permanent bearing temp. Upto 80° c	
	Speed upto 1500 rpm	Speed upto 1500 rpm
ESSO	ESSTIC 55 TERESSO 56	ESSIC 50 TERESSO 52
MOBILE	MOBILE VACTRA Oil heavy MOBILE D.T.E. Oil heavy VAC HLP 49	MOBILE VACTRA Oil heavy MEDIUM MOBILE D.T.E Oil heavy MEDIUM VAC HLP 36
SHELL	Shell Vitrea Oil 33 Shell Tellus Oil 33	Shell Vitrea oil 31 shell Tellus oil 29
VALVOLINE	VALVOLINE R-306	VALVOLINE R-206
AREL	AREL Oil CMY AREL Oil TU 7	AREL Oil CMY 7524 AREL Oil TU518
BP	BP ENERGOL HP 30	BP ENERGOL HP 20
IOC	SERVOSYSTEM 321	SERVOSYSTEM 317
CHEVRON	CHEVRON GP OIL 19 CHAVRON OC TURBINE OIL 19	CHEVRON GP OIL 14 CHEVRON OC TURBINE OIL 15
DEA	DEA VISCIBIL Seremit 6	DEA VISCIBIL Seremit 4

4.4 shafts

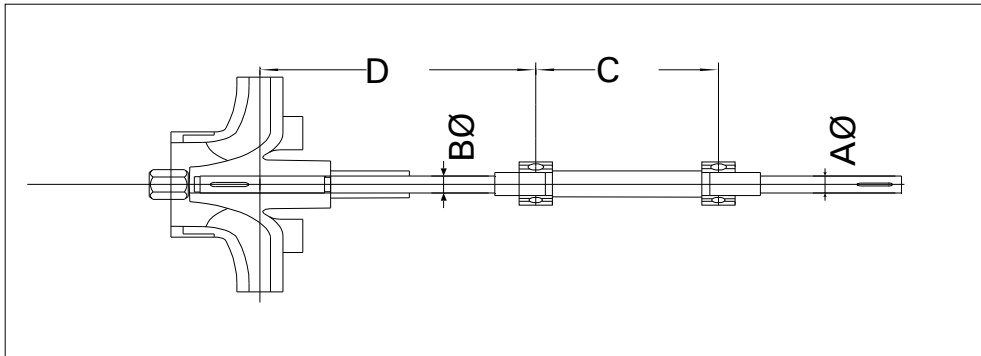
Only three shaft units are used for complete range of 'DB' pumps. These are made of high tensile steel and accurately ground all over.

Impeller Nominal Dia.(cm)	Delivery branch (mm)							
	32	40	50	65	80	100	125	150
13	Bearing Bracket 25/13							
16	Bearing Bracket 25/16							
20	Bearing Bracket 25/13							
26	Bearing Bracket 25/16				Bearing Bracket 35			
32							Bearing Bracket 45	
40								

4.4.1 Shaft transmission ratings for shaft of mat CS IS 1570 Gr 45C8

Shaft unit	at 1450 rpm	AT 2900 rpm
25	16 kw	33kw
35	49 kw	98 kw
45	82 kw	-

4.4.2 Shaft details



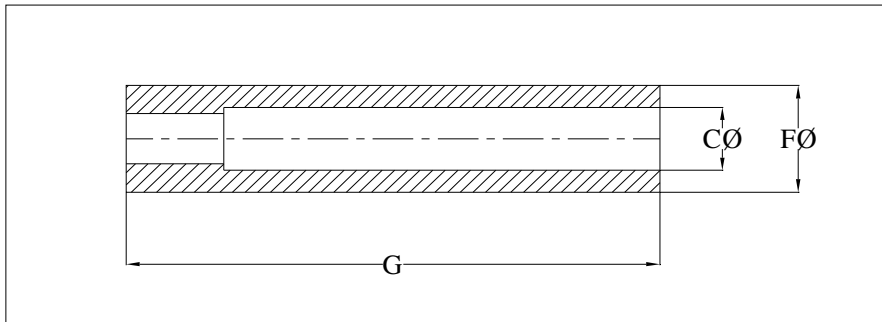
Shaft unit no.	A Dia. at coupling In mm	B Dia. under sleeve / stuffing box in mm	C Dist. bet. bearing in mm	D Dist. bet. NDE bearing and impeller in mm
25	24	24	117.3	165.3
35	32	32	161.3	194.3
45	42	44	162	221.5

4.4.3 Bearing brackets

For the three shaft units of 'DB' type pumps, there are four bearing brackets as shown in the table no. 4.4. On pumps with impellers of nominal diameter of 130 mm and 160 mm, the casing cover is clamped between the bearing bracket and the volute casing. On all other sides, except 80/20 and 100/20 the casing cover bolts on to the bearing bracket and the casing cover bolts on to the bearing bracket and the volute casing is bolted to the casing cover separately.

4.4.4 Shaft sleeve details

Shaft unit no.	F Outside diameter in mm	C inside diameter in mm	G Length in mm
25	32	24	75
35	40	32	85
45	55	44	90



Shaft Unit no	Depth In mm	Bore in mm	Size of packing in mm	Length of packing in mm	No. of packings with one lantern ring	Position from impeller side	Mechanical seal
25	50	48	8 sq	510	4	1 + L + 3	YES
35	63	60	10 sq	650	4	1 + L + 3	YES
45	63	75	10 sq	840	4	1 + L + 3	YES

4.5 Drive

4.5.1 Direct drive

'DB' pumps are directly driven by electric motor or engine. They use love- joy coupling L or C type of ample size as a standard arrangement. Other types of couplings can also be offered against specific requirement.

4.5.2 Indirect drive

Indirect drive is possible either via flat or v belts drive (refer to our sales head office)

4.5.3 Direction of rotation

'DB' pumps are assembled for clockwise rotation when viewed from coupling end. Reverse direction of rotation is not possible.

4.6 Coupling

4.6.1 Coupling details

Refer coupling selection chart no. CT 102-007-4

Chart No. : CT102 – 007 - 4

LOVE JOY TYPE COUPLING SELECTION FOR DB & DBE PUMPS.

- 1) Select the coupling as per H. P. & speed ratings from the chart of specific driving unit.
- 2) For the respective H.P. & rpm selected coupling shall be applicable for the frame size mentioned against the respective coupling.

3) For selection of coupling for frame sizes other than mentioned in the chart use following procedure.

- From H.P. and rpm, ratings find out the coupling from the chart for specific driving unit.
- See that shaft Dia. of motor does not exceed the max bore dia. specified for the selected coupling.
- If shaft dia. exceeds the max recommended bore for coupling select the next higher coupling.

Unit no 25 pump sizes	Coupling type			Max H P covered		Frame size covered		Max bore Φ in mm
	Stand.	Spacer	Spacer Length in mm	By motor		4pole	2pole	
				4 pole	2pole			
32/13 32/16 32/20 32/26	L-100	RRL-100	100	7.5 (5.5kw)	10 (7.5kW)	21 * 80 * 90 S 90 L 100 L 12 M	71 * 80 * 90 S 90L 100L 112M	35
40/13 40/16						132S	132S 132M	
40/20 40/26	L-110	RRL110 ▲	100	15 (11kW)	20 (15kW)	132M 160M	160M 160L	42
50/16 50/16 50/20	L-150	RRL150 ▲	100	—	30 (22kW)	—		48
50/26 65/13 65/16 65/20	L-190	RRL190 ▲	100	—	60 (45kW)	—		55
80/16	L-225	RRL225 ▲	100	—	60 (45kW)	—	200 M 200 L	60

Note 1) ▲: - Pump side coupling triangular.

2) *:- Motor side coupling should be as per Drg. No. M90631

Unit no 35 pump sizes	Coupling type			Max. H. P. covered By motor		2 Pole	Frame size covered			Max bore Φ In mm
	Standard	Spacer	Spacer Length in mm	6 pole	4pole		6 Pole	4Pole	2pole	
65/26 65/32 80/20	L-100	RRL-100	140	–	7.5 (5.5Kw)	10 (7.5kw)	–	KH 90S KH 90L KH 100L KH112M	KH 90S KH 90L KH 100L KH112M	35
80/26 80/32	L-110	RRL110	140	10 (7.5KW)	15 (11KW)	20 (15KW)	KH 90S/L KH100L KH112M KH132S/ M	KH 132S KH132M KH160M	KH132S KH132M KH160M KH160L	42
100/20 100/26 100/32	L-150	RRL110	140	15 (11KW)	20 (15KW)	30 (22KW)	LC160M LC160L KH 160L	LC160M LC160L LD180M KH160L	LC160M LC160L LD180M KH160L	48
125/26	L-190	RRL150	140	–	30 (22KW)	60 (45KW)	–	LC180M LD180L	LC180M LC180L LD200L LC225S KH250 M	55
	L-225	RRL225 ▲	140	–	40 (30KW)	75 (55KW)	–	LD200L	LC200M LC 200L LC225M LC250M	60
	C226	CQ 226	140	–	–	120 (90KW)	–	–	LC250S LC250M	60
	C-276	CQ 276	140	–	60 (45KW)	120 (11KW)	–	–	LC280S LC280M LD280S LD280M LD315S LD315M	75

Note 1) ▲ : - Pump side coupling triangular.

Unit No. 45 pump sizes	Coupling type			Max H P covered By motor		Frame size covered		Max bore Φ In mm
	Stand.	Spacer	Spacer Length in mm					
				6 pole	4pole	6 Pole	4Pole	
125/32	L-110	RRL-110	140	10 (7.5KW)	15 (11KW)	KH132S KH132M	KH 132S KH132M KH160M	42
125/40 100/40 150/32	L-150	RRL150	140	15 (11KW)	25 (18.5KW)	KH 160M KH 160L LC160M LC160L	KH 132S KH132M KH160M	48
150/40	L-190	RRL190	140	20 (15KW)	30 (22KW)	LC180M LD180L	KH160L LC160M LC160L LD180M	55
	L-225	RRL225	140	40 (30KW)	50 (37KW)	LD200L LC180L LC200M	LC180M LD180L	60
	C-226	CQ226	140	—	60 (45KW)	LC200L LD225M	LC200L LD225M	60
	C276	CQ 276	140	—	100 (75KW)	-	LC225M LC250M LD250M LD280S	75
	C-280	CQ 280	140	—	120 (90KW)	—	LC250M LD280M	75

T NO. CM102-005-2

PART CODE		MATERIAL CONSTRUCTION CHART													KEY TO MATERIAL CODE	
		16	34	35	48	56	57	58	77	78	83	84	86	87		
05	PUMP CASING	012	012	012	012	012	012	012	112	110	012	012	012	012	MAT	
51	IMPELLER	012	112	112	110	110	110	110	112	110	012	110	110	110	012	CI.IS.210 GR FG 260
80	SHAFT	253	056	253	056	253	056	253	253	253	056	056	056	253	052	CS.IS1570 GR 20 C8
90	CASE WEAR RING	012	110	110	110	012	110	110	110	110	012	110	012	110	053	CS.IS1570 GR 40 C8
20	CASING COVER	012	012	012	012	012	012	012	112	110	012	012	012	012	056	CS.IS1570 GR 45 C8
23	GLAND	012	012	012	012	012	012	012	110	110	012	110	012	110	110	BR IS 318 GR LTB 2
27	LATERN RING	364	012	012	364	364	364	364	110	110	364	110	364	110	112	PH BR IS 28 GR 1
10	SHAFT SLEVE	565	565	565	110	565	110	110	110	110	565	110	565	110	253	ST ST ASTM A 276 TYPE 410
20	IMPELLER KEY	253	053	253	053	253	053	253	253	253	053	053	053	253	272	IS 1367 CL 6.8
30	IMPELLER NUT	253	277	253	253	253	277	253	253	253	277	277	277	253	277	IS 1367 CL 6
81, 82	NUTS	278	278	278	278	278	278	278	253	253	278	278	278	278	278	IS 1367 CL 8
90, 91	STUDS	272	272	272	272	272	272	272	253	253	272	272	272	272	565	ST ASTM A 276 TYPE 410 H (210 TO 250 BHN)
00	PLUG IN LIQUID CONTACT	052	052	052	052	052	052	052	253	253	052	052	052	052	364	NYLON

Note: Rest of the parts are common as per material construction code - 83.

5.1 Painting

The details of paint used are as follows:

Paint type: special synthetic enamel fast drying

Shade: violet blue RAL500

Dry film thickness: 50 micron

6 Design criteria

6.1 Viscosity

‘DB’ pumps can be used for liquid having viscosity upto 500 SSU, However viscosity correction is to be applied for viscosity above 55 SSU.

6.2 Reserve power margin. The motor should be selected with a sufficient reserve margin. The relation between the reserve power and absorbed by the pump is as under.

Power absorbed in kW	reserve power margin in kW
Upto 1.5	at least 0.75 kW

From 1.5 to 15	15% approx
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Above 15	10% approx
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6.3 Moment of inertia

The moment of inertia of complete rotating unit without coupling but with full size impeller and filled with water are as under in kg.cm²

Size of the impeller	M.I. impeller of		Dyn. moment of inertia impeller of	
	C.I.	LT Br IS 318 GR LTB 2	C.I.	LT Br IS 318 GR LTB 2
32/13	110	120	3	3
32/16	330	380	8	10
32/20	730	830	19	2
32/26	1780	2020	45	151
40/13	170	190	4	5
40/16	320	360	8	9
40/20	750	860	19	22
40/26	1870	2150	48	55
50/13	200	220	5	6
50/16	390	430	10	11
50/20	810	930	20	24
50/26	2100	2500	56	64
50/32K	4940	5830	126	146
65/13	260	290	7	7
65/16	480	550	12	15
65/20	1070	1200	27	31
65/26	2410	2700	61	69
65/32	4940	5830	126	146
80/16	800	660	15	17
80/20	1170	1310	30	33
80/26	2610	2930	67	74
80/32	4720	5370	120	137
100/20	1450	1600	37	41
100/26	3180	3550	81	90

100/32	6510	7350	166	187
100/40	15900	18120	451	462
125/26	3190	3570	81	91
125/32	7120	8030	181	205
125/40	18260	20240	465	516
150/32	7680	8600	196	219
150/40	17750	19910	452	507

6.3.1 M.I. OF IMPELLERS OTHER THAN C.I.

If impellers are made of other than C.I. the moment of inertia can be found out by using following formula

$$\text{M.I. of impeller in Kg-cm}^2 = \frac{(\text{Sp. gr. of impeller material})}{\text{Sp. Gr. of CI}} \times \text{M.I. of CI diameter}$$

6.3.2 MOMENT OF INERTIA OF CUTDOWN IMPELLER

If the moment of inertia of cut down impeller diameters is to be calculated, it can be obtained by using the following formula

$$\text{M.I. of cut down Dia impellers} = \left\{ \frac{\text{cut down dia}}{\text{full dia}} \right\}^{2.5} \times \text{M.I. full dia impeller}$$

$$6.3.3 \text{ GD2 VALUE} = \text{M I} \times 4$$

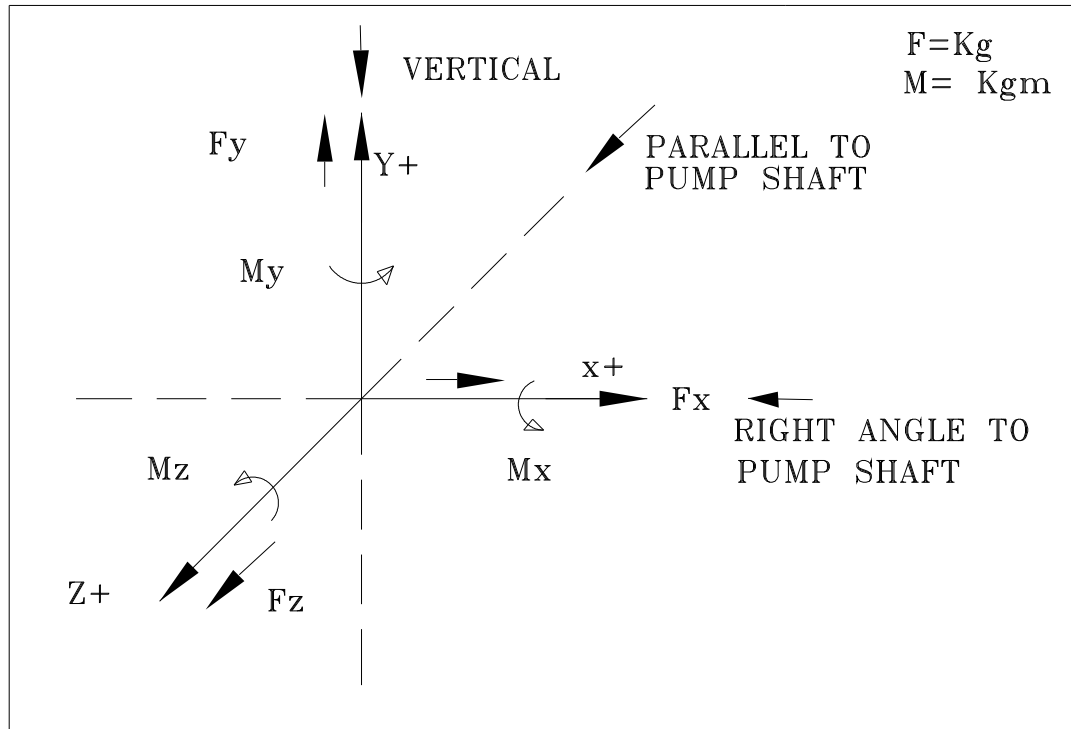
6.4 WEIGHTS

The net weights of pumps with different material of construction are as follows

Size	Total net weight of pump having volute casing and casing cover of	
	C.I.IS 210 GR FG 260 in kg	LT Br IS 318 GR LTB 2
32/13	28	32
32/16	34	39
32/20	41	48
32/26	59	70
40/13	29	33
40/16	36	41
40/20	44	52
40/26	61	72
50/13	32	37
50/16	38	43
50/20	46	54
50/26	63	74
50/32K	90	96
65/13	39	45
65/16	43	50
65/20	52	61
65/26	81	94
65/32*	100	117
80/16	53	62
80/20	70	80
80/26	91	106
80/32	107	126
100/20	85	99
100/26	103	122
100/32	116	137
100/40	177	209
125/26	110	129
125/32	151	177
125/40	188	223
150/32	170	200
150/40	205	242

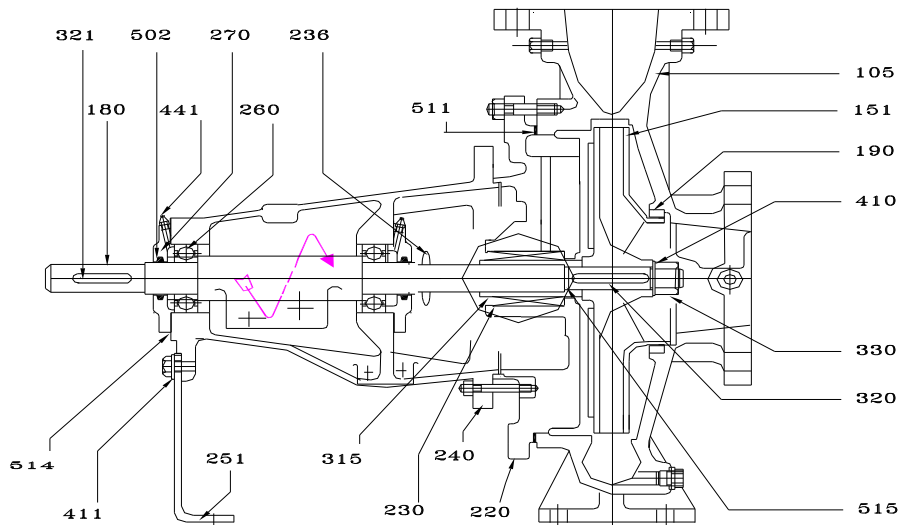
6.5 Allowable forces & moment on pump flanges

Allowable unbalanced forces and moments as per NEMA STANDARDS on suction and delivery flanges.



SIZE	$\pm F_x$	$\pm F_y$	$\pm F_z$	$\pm M_x$	$\pm M_y$	$\pm M_z$
32mm	28	71	55	43	22	22
40mm	34	85	68	52	26	26
50mm	45	114	91	69	35	35
65mm	57	142	114	86	43	43
80mm	68	170	136	104	52	52
100mm	91	227	182	138	69	69
125mm	114	284	227	173	86	85
150mm	136	340	272	208	104	104
200mm	182	454	364	276	138	138

CROSS SECTIONAL ASSEMBLY

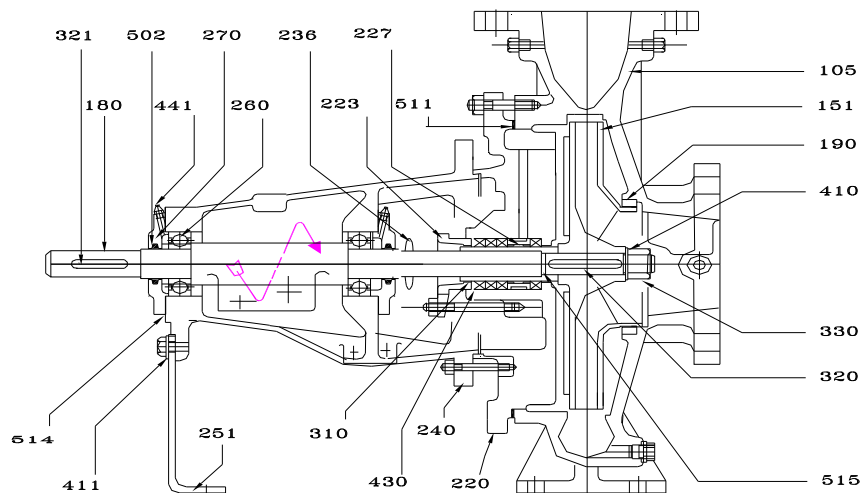


32/13 TO 65/13
32/16 TO 80/16
32/20 TO 100/20
32/26 TO 125/26

SR NO	PART NO.	DESCRIPTION	MATERIAL SPECIFICATION	QTY
1	105	PUMP CASING	CI IS 210 GR FG 260	1
2	151*	IMPELLER	CI IS 210 GR FG 260	1
3	180*	SHAFT	CS IS 1570 GR 45 C8	1
4	190*	CASING RING	CI IS 210 GR FG 260	1
5	220	CASING COVER FOR MECH SEAL	CI IS 210 GR FG 260	1
6	223	GLAND	CI IS 210 GR FG 260	1
7	227	LATERN RING	CI IS 210 GR FG 260	1
8	310*	SHAFT SLEEVE	ST ST ASTM A276 TYPE 410H	1
9	320*	IMPELLER KEY	CS IS 1570 GR-40 C8	
10	330	IMPELLER NUT	IS 1367 P6 CL-8	1
11	260*	BALL BRG. SKF 6305 OR EQ.	BRG. STEEL	2
12	270	BEARING COVER	CI IS 210 GR FG 200	1
13	236*	LIQUID DEFLECTOR	RUBBER	1
14	240	BEARING HOUSING	CI IS 210 GR FG 200	1
15	321	COUPLING KEY	CS IS 1570 GR 40 C8	1
16	410*	SUPPORT FOOT	MS IS 1079 GR ST 34	1
17	411	LOCK WASHER	CS IS 4072 GR 80 C6	1
18	411	LOCK WASHER	CS IS 4072 GR 80 C6	2
19	411	GLAND PACKING	GRAPHITED CARBAN	2
20	441	GREASE NIPPLE	IS 1367 CL 4.6	1
21	502*	FELT RING	FELT	2
22	511*	GASKET FOR S.NO.1	NON ASBESTOSE -FERROLITE NAM37	1
24	515*	GASKET FOR S.NO.12	NON ASBESTOSE -FERROLITE NAM37	2
24	515*	GASKET FOR S.NO.8	NON ASBESTOSE -FERROLITE NAM37	1

* RECOMMENDED SPARES

CROSS SECTIONAL ASSEMBLY



65/ 32 TO 150 / 32
100/ 40 TO 150/ 40

SR NO	PART NO.	DESCRIPTION	MATERIAL SPECIFICATION	QTY
1	105	PUMP CASING	CI IS 210 GR FG 260	1
2	151*	IMPELLER	CI IS 210 GR FG 260	1
3	180*	SHAFT	CS IS 1570 GR 45 C8	1
4	190*	CASING RING	CI IS 210 GR FG 260	1
5	220	CASING COVER FOR MECH SEAL	CI IS 210 GR FG 260	1
6	223	GLAND	CI IS 210 GR FG 260	1
7	227	LATERN RING	CI IS 210 GR FG 260	1
8	310*	SHAFT SLEEVE	ST ST ASTM A276 TYPE 410H	1
9	320*	IMPELLER KEY	CS IS 1570 GR-40 C8	
10	330	IMPELLER NUT	IS 1367 P6 CL-8	1
11	260*	BALL BRG. SKF 6305 OR EQ.	BRG. STEEL	2
12	270	BEARING COVER	CI IS 210 GR FG 200	1
13	236*	LIQUID DEFLECTOR	RUBBER	1
14	240	BEARING HOUSING	CI IS 210 GR FG 200	1
15	321	COUPLING KEY	CS IS 1570 GR 40 C8	1
16	410*	SUPPORT FOOT	MS IS 1079 GR ST 34	1
17	411	LOCK WASHER	CS IS 4072 GR 80 C6	1
18	411	LOCK WASHER	CS IS 4072 GR 80 C6	2
19	411	GLAND PACKING	GRAPHITED CARBAN	2
20	441	GREASE NIPPLE	IS 1367 CL 4.6	1
21	502*	FELT RING	FELT	2
22	511*	GASKET FOR S.NO.1	NON ASBESTOSE -FERROLITE NAM37	1
24	515*	GASKET FOR S.NO.12	NON ASBESTOSE -FERROLITE NAM37	2
24	515*	GASKET FOR S.NO.8	NON ASBESTOSE -FERROLITE NAM37	1
25	190.2*	CASING RING (DEL. SIDE)	CI IS 210 GR FG 260	1

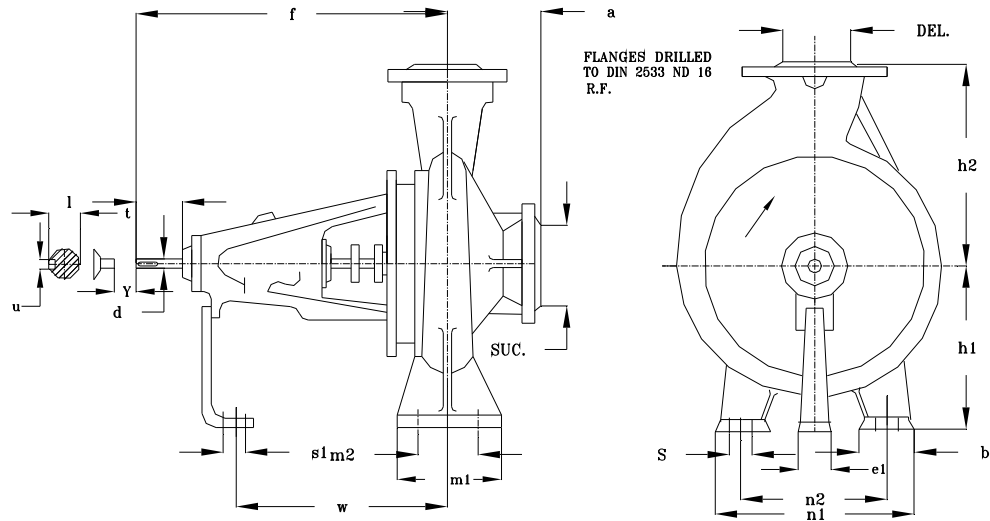
* RECOMMENDED SPARES

DRG. NO.

KIRLOSKAR BROTHERS LIMITED

TL10294001-0

OUTLINE DRAWING OF KIRLOSKAR DB PUMPS.



SIZE	UNIT	PUMP DIMENSIONS						FOOT DIMENSIONS										SHAFT END				Y
		DEL.	SUC.	a	f	h1	h2	b	c	m1	m2	n1	n2	s	e1	s1	w	d	l	t	u	
32/13	25	32	50	80	360	112	140	50	14	100	70	190	140	14	100	14	267	24	50	27	8	100
32/16	25			80	360	132	160	50	14	100	70	240	190	14	100	14	267	24	50	27	8	100
32/20	25			80	360	160	180	50	14	100	70	240	190	14	110	14	267	24	50	27	8	100
32/26	25			100	360	180	225	65	14	125	95	320	250	14	110	14	267	24	50	27	8	100
40/13	25	40	65	80	360	112	140	50	14	100	70	210	160	14	100	14	267	24	50	27	8	100
40/16	25			80	360	132	160	50	14	100	70	240	190	14	100	14	267	24	50	27	8	100
40/20	25			100	360	160	180	50	14	100	70	265	212	14	110	14	267	24	50	27	8	100
40/26	25			100	360	180	225	65	14	125	95	320	250	14	110	14	267	24	50	27	8	100
50/13	25	50	65	100	360	132	160	50	14	100	70	240	190	14	110	14	267	24	50	27	8	100
50/16	25			100	360	160	180	50	14	100	70	265	212	14	110	14	267	24	50	27	8	100
50/20	25			100	360	160	200	50	14	100	70	265	212	14	110	14	267	24	50	27	8	100
50/26	25			100	360	180	225	65	14	125	95	320	250	14	110	14	267	24	50	27	8	100
65/13	25	65	80	100	360	160	1800	65	14	125	95	280	212	14	110	14	267	24	50	27	8	100
65/16	25			100	360	160	200	65	14	125	95	280	212	14	110	14	267	24	50	27	8	100
65/20	25			100	360	180	225	65	14	125	95	320	250	14	110	14	267	24	50	27	8	100
65/26	35			100	470	200	250	80	16	160	120	360	280	18	110	14	342	32	80	25	10	140
65/32	35	80	100	125	470	225	280	80	16	160	120	400	315	18	110	14	342	32	80	35	10	140
80/16	25			125	470	180	225	65	14	125	95	320	250	14	110	14	267	24	50	27	8	100
80/20	35			125	470	180	250	65	14	125	95	345	280	14	110	14	267	32	80	35	10	140
80/26	35			125	470	200	280	80	16	160	120	400	315	18	110	14	267	32	80	35	10	140
80/32	35	100	125	125	470	250	315	80	16	160	120	400	315	18	110	14	342	32	80	35	10	140
100/20	35			125	470	200	280	80	16	160	120	360	280	18	110	14	342	32	80	35	10	140
100/26	35			140	470	225	280	80	16	160	120	400	315	18	110	14	342	32	80	35	10	140
100/32	35			140	470	250	315	80	16	160	120	400	315	18	110	14	342	32	80	35	10	140
100/40	45	125	150	140	530	180	355	100	18	200	150	500	400	23	110	14	370	42	110	45	12	140
125/26	35			140	470	250	355	80	16	160	120	400	315	18	110	14	342	32	80	35	10	140
125/32	45			140	530	280	355	100	18	200	150	500	400	23	110	14	370	42	110	45	12	140
125/40	45			140	530	315	400	100	18	200	150	500	400	23	110	14	370	42	110	45	12	140
150/32	45	150	200	160	530	280	400	100	18	200	150	550	450	23	110	14	370	42	110	45	12	140
150/40	45			160	530	315	450	100	18	200	150	550	450	23	110	14	370	42	110	45	12	140
50/32K	35	50	65	125	470	225	290	80	16	160	120	400	315	18	110	14	342	32	80	35	10	140